The falling chain: the scale reads THREE times the landed weight

Three-times-weight derivation; corrected OL1/D7.10 bookkeeping; variants; problem set.

The falling chain: the scale reads THREE times the landed weight

Setup: a uniform chain of mass m, length L, linear density $\lambda = m/L$ is released onto a scale (either suspended with its lower end just touching the pan, or lifted as a strand and dropped - the two readings agree, see below). Let x be the length that has LANDED (equal to the distance the chain has fallen). The airborne part moves at $v = \sqrt{2gx}$;

mass arrives at rate $\dot{m} = \lambda v = \lambda\sqrt{2gx}$; the impact force stopping it is $\dot{m}v = \lambda v^2 = 2\lambda g x$. The scale also carries the landed weight $\lambda x g$. Total: $F = \lambda x g + 2\lambda g x = 3\lambda x g$ - THREE times the weight already on the pan.

Falling chain: scale reads THREE times landed weight (cont.)

At the instant the last link lands ($x = L$): $F = 3\lambda L g = 3mg$ - the famous three-times-weight reading. Limits: as $x \to 0$, $F \to 0$ (nothing on the pan); the reading climbs linearly with x; the peak is exactly at the completion of the landing.

If instead the chain is dropped from a height h ABOVE the pan in a heap (all links fall together), the reading is a single impulse $J = m\sqrt{2gh}$ over the collision time, not the smooth $3mg$ ramp - geometry decides the history, but the flux law $\dot{m}v$ never changes.

Falling chain: scale reads THREE times landed weight (cont.)

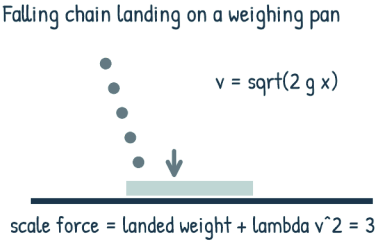
Correction to the source (recorded in SOURCE-AUDIT): source OL1 and its D7.10 brief compute $F = \lambda(L + x)g$ with x = length still falling and conclude $2mg$ at the end. That bookkeeping pairs the wrong speed with the wrong length (it sets $v = \sqrt{2g\ell}$ with ℓ the AIRBORNE length; in fact the airborne speed is $\sqrt{2g \times \text{descent}} = \sqrt{2gx_{landed}}$). The source README itself advertises the 'three-times-weight result' - the corrected derivation delivers exactly that. Use $F = 3\lambda x g$ with x = landed length = descent.

Falling chain: scale reads THREE times landed weight (cont.)

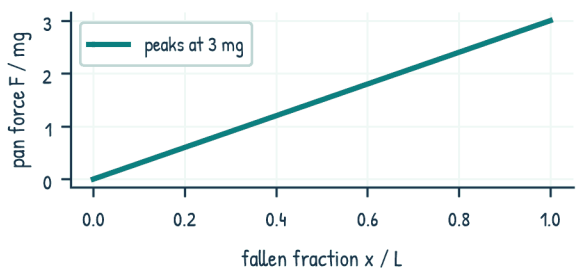
WHY / the factor 3 = 1 (weight) + 2 (flux): the flux term $\lambda v^2 = 2\lambda g x$ is TWICE the landed weight $\lambda x g$ at every instant because $v^2 = 2gx$ while weight grows as gx. Momentum flux is not a small correction here - it dominates.

CHECK / units: $3\lambda x g = 3$ (kg/m)(m)(m/s^2) = N. Derive the flux term independently: momentum of the falling part changes as links are arrested; force = rate of momentum destruction $= (\lambda v)v$.

Falling chain links hit pan at sqrt(2gx), adding 2 lambda g x flux



Scale reading climbs linearly to 3 mg right as the top link lands



REMEMBER / Falling chain: scale reads three times the landed weight, ending at 3mg; momentum flux plus weight, no exceptions.

The falling chain: the scale reads THREE times the landed weight (cont.)

Three-times-weight derivation; corrected OL1/D7.10 bookkeeping; variants; problem set.

Variant geometry: the strand lifted from a pile and released

Variant B (source OL1's words 'piled on a scale, one end lifted and released'): lift a strand (the whole chain) and release. The falling portion descends with $v^2 = 2gy$ where y is the DESCENT (not the airborne length!). Landing rate λv, impact $\lambda v^2 = 2\lambda g y$; landed weight $\lambda y g$ (the landed length equals y when the strand's lower end started at the pan).

Then $F = 3\lambda y g$ again - identical formula with y = descent = landed length. The 3mg endpoint is robust across these geometries.

Variant geometry: the strand lifted from a pile and released (cont.)

If instead the whole chain is a heap at rest and simply released from height H above the pan (all links airborne at once): the reading is an IMPULSE spike $J = m\sqrt{2gH}$ averaged over the short collision, not a smooth ramp - different problem, same momentum bookkeeping. The exam rule: identify what is falling, at what speed, landing at what rate - then $F =$ landed weight $+\dot{m}v$.

CHECK / dimensional and limit checks on variant B: $F(y) = 3\lambda y g$ grows from 0 to 3mg; quasi-static limit: at vanishing fall speed the flux term dies and the pan reads just the landed weight.

Why the naive answer is 2mg (and why it fails)

The tempting wrong bookkeeping: airborne length ℓ, speed $\sqrt{2g\ell}$ (WRONG pairing: speed follows the FALL DISTANCE, not the remainina airborne lenath), landed $= L - \ell$, giving $F = \lambda(L-\ell)g + 2\lambda g\ell = \lambda(L+\ell)g \to 2mg$.

The error:

for a strand whose leading end has descended y, the airborne portion has length $L - y$ (started as one full strand) and speed $\sqrt{2gy}$; substituting $\ell = L - y$ into $\sqrt{2g\ell}$ mis-states the speed whenever $y \neq \ell$. Only at ONE instant ($y = \ell = L/2$) do the two agree - which is why the wrong answer survives spot checks at mid-fall.

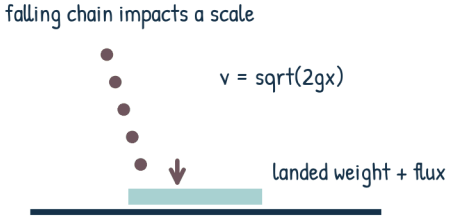
Why the naive answer is 2mg (and why it fails) (cont.)

Sanity probe that exposes it: at $y = L/2$ both give $F = 1.5mg$ (wrong: $\lambda(L + L/2)g = 1.5mg$; right: $3\lambda(L/2)g = 1.5mg$); at $y = L$: wrong $2mg$ vs right $3mg$. The ENDPOINT distinguishes them; the 3mg endpoint is the published classic.

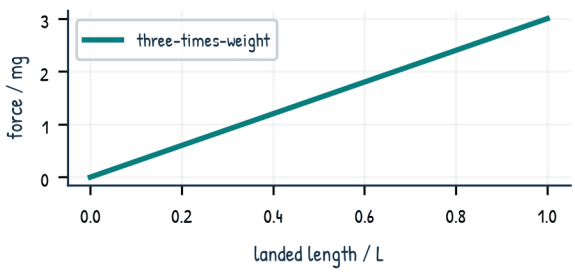
WHY / the invariant rule: $F = W_{landed} + \dot{m}v$ with $\dot{m} = \lambda v$ and v the ACTUAL landing speed measured from the fall distance. One diagram with labelled speeds prevents the pairing error.

CHECK / the same 3mg endpoint appears in the equivalent 'chain poured onto a pan at free-fall speed' limit and in the Irodov-style textbook solutions - consistent across treatments.

falling chain: landed length x, speed and flux



chain three: limiting curve / diagnostic



REMEMBER / Falling chain: scale reads three times the landed weight, ending at 3mg; momentum flux plus weight, no exceptions.

The falling chain: the scale reads THREE times the landed weight (cont.)

Three-times-weight derivation; corrected OL1/D7.10 bookkeeping; variants; problem set.

Chain-flux problem set

(a) Chain 0.5 kg/m falling onto a pan; reading when 0.4 m has landed (g = 10): $F = 3(0.5)(0.4)(10) = 6$ N. (b) At completion ($L = 2$ m, $m = 1$ kg): $F = 3(1)(10) = 30$ N. (c) Same chain held with the end just on the pan and released - same formula. (d) Compare: the same chain RESTING on the pan reads $mg = 10$ N; the falling version peaks at 3x.

(e) Sand falling at constant rate at constant speed (CM-16) reads $W + \dot{m}v$; the chain adds the $v^2 = 2gx$ growth. (f) The chain LIFTED at constant speed (CM-18) reads $\lambda x g + \lambda v^2$ with v = lift speed (constant) - the weight term grows, the flux term is constant; the falling chain has BOTH growing.

Chain-flux problem set (cont.)

CHECK / drill (a): $3\lambda x g = 3 \times 0.5 \times 0.4 \times 10 = 6$ N; landed weight 2 N; flux 4 N $= 2\lambda g x$ - the 1:2 split.

Setup: projectile m_1 at speed u hits target m_2 at rest; elastic; smooth; deflects at lab angle θ_1 with target recoil θ_2. Three regimes. (1) $m_1 < m_2$: the projectile can scatter BACKWARD; θ_1 ranges over the full $(0, 180)$ degrees - e.g. a neutron off a lead nucleus bounces anywhere. (2) $m_1 = m_2$: $\theta_1 + \theta_2 = 90$ degrees exactly (CM-12); θ_1 itself spans $(0, 90)$.

Scattering limits: what mass ratios do to the exit angles (cont.)

(3) $m_1 > m_2$: the projectile is FORCED FORWARD: θ_1 is capped at $\theta_{max} = \arcsin(m_2/m_1)$ (the circle-locus picture of CM-15: $\mathbf{v}_1 = \mathbf{v}_{cm} + \mathbf{v}'_1$ is the sum of a fixed vector and a vector of fixed length; the extreme direction is tangent to that circle). A bowling ball cannot bounce back off a marble - at best it scatters by $\arcsin(m_2/m_1)$.

The lab target angle: $$\tan\theta_2 = \frac{\sin\theta_{cm}}{\cos\theta_{cm} + m_1/m_2}$$

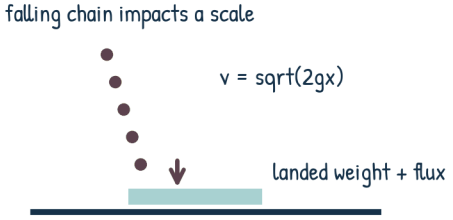
where θ_{cm} is the CM-frame scattering angle of the projectile (source OL2's final formula - correct as printed). Consequences: $\theta_2 = (180 - \theta_{cm})/2$ when $m_1 = m_2$; $\theta_2 \le 90$ degrees always;

Scattering limits: what mass ratios do to the exit angles (cont.)

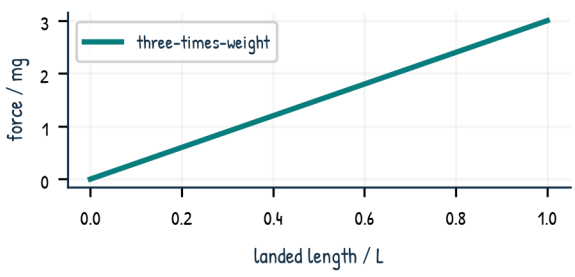
in the extreme-mass limits $\theta_2 \to \theta_{cm}$ (heavy target) or $\theta_2 \to 0$ (light target).

Correction to the source (recorded in SOURCE-AUDIT): source OL2 lists '$\theta_{max} = \arcsin(m_2/m_1)$ if $m_1 < m_2$' and 180/90 otherwise. The $\arcsin$ cap applies when the PROJECTILE IS HEAVIER ($m_1 > m_2$); for $m_1 < m_2$ there is no cap at all (θ_1 can reach 180). The intermediate values ($\theta_{max} = 180$ for $m_1 < m_2$; 90 for equality; $\arcsin(m_2/m_1)$ for $m_1 > m_2$) are the corrected table.

falling chain: landed length x, speed and flux



chain three: limiting curve / diagnostic



REMEMBER / Falling chain: scale reads three times the landed weight, ending at 3mg; momentum flux plus weight, no exceptions.

Scattering limits: what mass ratios do to the exit angles

Corrected scattering-limit table; circle-locus geometry; tan(theta2) formula; drills.

Scattering limits: what mass ratios do to the exit angles (cont.)

WHY / the cap is the tangency of the momentum triangle: $\mathbf{v}_1 = \mathbf{v}_{cm} + \mathbf{v}'_1$ with $v_{cm} = \frac{m_1}{m_1+m_2}u$ and $v'_1 = \frac{m_2}{m_1+m_2}u$ (CM speeds). The direction of $\mathbf{v}_1$ rotates as the impact parameter varies; θ_1 extremises when $\mathbf{v}_1$ is tangent to the circle: $\sin\theta_{max} = v'_1/v_{cm} = m_2/m_1$ - exactly the condition $m_2 \le m_1$.

CHECK / limit $m_2 \to m_1$: $\arcsin(1) = 90$ degrees - the 90-degree theorem's boundary. $m_2 \ll m_1$: $\theta_{max} \to 0$: heavy projectile never deflects. $m_2 > m_1$: the tangent construction fails and every $\theta_1 \in (0, 180)$ is reachable.

Worked: the maximum-deflection geometry

Head-on elastic against a target of mass $m_2 < m_1$ would send $v_1 = \frac{m_1-m_2}{m_1+m_2}u > 0$ straight through; the MAXIMUM deflection instead occurs at a glancing impact parameter. Method: (1) compute $v_{cm} = \frac{m_1 u}{m_1+m_2}$, CM speeds $u'_1 = \frac{m_2 u}{m_1+m_2}$, $u'_2 = -\frac{m_1 u}{m_1+m_2}$;

(2) elastic CM reversal rotates the relative velocity but preserves each CM speed; (3) $\mathbf{v}_1 = \mathbf{v}_{cm} + \mathbf{v}'_1$ traces the circle radius u'_1 centred at v_{cm}; (4) the deflection angle is extremal at tangency: $\sin\theta_{max} = u'_1/v_{cm} = m_2/m_1$; (5) the speed at the tangent point: $v_1 = \sqrt{v_{cm}^2 - u_1'^2} = u\frac{\sqrt{m_1^2 - m_2^2}}{m_1+m_2}$.

Worked: the maximum-deflection geometry (cont.)

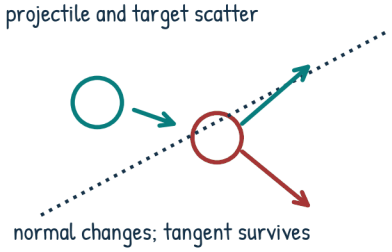
Numbers: $m_1 = 2$, $m_2 = 1$, $u = 6$: $v_{cm} = 4$, $u'_1 = 2$; $\sin\theta_{max} = 0.5$: $\theta_{max} = 30$ degrees; $v_1 = \sqrt{16-4} = 3.46$ m/s at that extremum; target flies off at θ_2 with speed set by the momentum triangle: $\mathbf{v}_2 = \mathbf{v}_{cm} + \mathbf{v}'_2$ with $|v'_2| = 4$;

compute θ_{cm} from $\tan\theta_1 = \frac{u'_1\sin\theta_{cm}}{v_{cm} + u'_1\cos\theta_{cm}}$ (projectile lab angle from the CM decomposition).

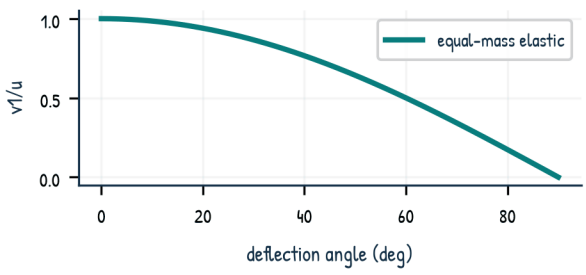
Worked: the maximum-deflection geometry (cont.)

CHECK / verify the 2:1 numbers via the master formulas at a special impact parameter: head-on $\theta = 0$: $v_1 = (2-1)6/3 = 2$, $v_2 = 2(2)(6)/3 = 8$ - circle check: $v_1 = |v_{cm} - u'_1| = 2$; opposite extreme $v_1 = v_{cm} + u'_1 = 6$ (zero deflection) - the circle intersects the forward axis at 2 and 6; the deflection extremum is the tangent at 30 degrees - consistent.

scatter: labelled physical setup



scatter: limiting curve / diagnostic



REMEMBER / Lab angle: tan(theta) = sin(Theta)/(cos(Theta) + m1/m2); heavy projectiles cannot exceed arcsin(m2/m1).

Scattering limits: what mass ratios do to the exit angles (cont.)

Corrected scattering-limit table; circle-locus geometry; tan(theta2) formula; drills.

Olympiad problem patterns (source OL2 family)

'Find the maximum angle the projectile can be deflected through' -> $\arcsin(m_2/m_1)$ if heavier, else 180 degrees. 'Find θ_2 in terms of θ_{cm} and masses' -> the $\tan\theta_2$ formula. 'Show $\theta_1+\theta_2=90$ for equal masses' -> CM-12's dot product. 'Find the impact parameter for maximum deflection' -> the tangent geometry: $b=(R_1+R_2)\sin\phi$ with ϕ set by the tangent construction - derive per problem from the circle.

Olympiad problem patterns (source OL2 family) (cont.)

Particle-physics dressing: same math as alpha-particle scattering at fixed energy; the $\arcsin(m_2/m_1)$ cap is why fast neutrons thermalise efficiently off hydrogen (equal masses, big deflections) and can even backscatter off heavy nuclei ($m_1<m_2$: full range).

CHECK / restated limits table (the corrected version): $m_1<m_2$: θ_1 any angle to 180; $m_1=m_2$: $\theta_1+\theta_2=90$; $m_1>m_2$: $\theta_1\le\arcsin(m_2/m_1)$.

Memorise the table with its regime labels.

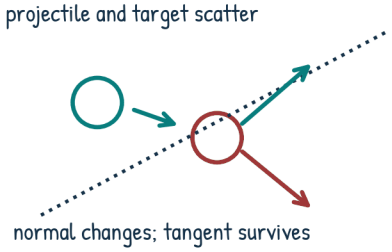
Drills on the circle locus

(a) $m_1=3$, $m_2=1$: $\theta_{max}=\arcsin(1/3)=19.5$ degrees. (b) $m_1=1$, $m_2=3$: projectile can rebound; the target recoils forward within $\theta_2\le 90$. (c) $m_1=m_2$, projectile deflected 53 degrees: $\theta_2=37$; $v_1=u\cos 53=0.6u$, $v_2=u\sin 53=0.8u$ (CM-12).

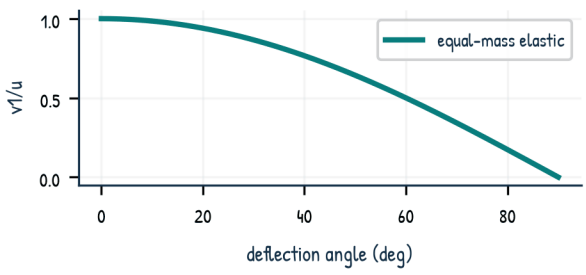
Drills on the circle locus (cont.)

(d) Energy check on the tangent point of the 2:1 example: $v_1=3.46$; $K_1=\frac{1}{2}(2)(12)=12$ J; target from energy: $m_2v_2^2=m_1(u^2-v_1^2)=2(36-12)=48$; $v_2=\sqrt{48}=6.93$; $K_2=\frac{1}{2}(1)(48)=24$ J; total 36 J $=K_i$.

scatter: labelled physical setup



scatter: limiting curve / diagnostic



REMEMBER / Lab angle: tan(theta) = sin(Theta)/(cos(Theta) + m1/m2); heavy projectiles cannot exceed arcsin(m2/m1).

Scattering limits: what mass ratios do to the exit angles (cont.)

Corrected scattering-limit table; circle-locus geometry; tan(theta2) formula; drills.

Drills on the circle locus (cont.)

CHECK / drill (d) closes the energy circle exactly - the tangent point is a legitimate elastic outcome.

A ball collides with a fixed surface over a short time τ with a contact force $F(t)$ that spikes far above every other force in the problem. The impulse approximation: ignore gravity/friction during τ and write $J = \int_0^\tau F dt = \Delta p$; then resume the smooth motion with the new velocity. Source OL3: 0.1 kg ball at 10 m/s bouncing elastically off a floor:

When a finite force acts for a blink: the impulse approximation (co.

$J = 2mv = 2$ N s (reversal doubles the momentum swing). If instead the contact force peaks at 1000 N over 2 ms: $J_{contact} = 2$ N s exactly - the pulse alone reverses the ball; gravity's impulse over 2 ms is $mg\tau = 0.002$ N s, a 0.1% correction. The approximation is honest: state the scale separation.

Source OL3's second half: same ball rolling at 10 m/s stopped by friction $\mu = 0.5$ in 2 s: impulse $= \mu mgt = 0.5(1)(2) = 1$ N s - NOT equal to the collision impulse 2 N s because the ball's momentum change is only 1 N s (stops, no reversal). The comparison teaches impulse bookkeeping: $J = |\Delta p|$ for whatever the actual velocity change is; reversal costs 2mv, stopping costs mv.

When a finite force acts for a blink: the impulse approximation (co.

Force-pulse problem types: (i) given $F(t)$ shape (rectangular, triangular, half-sine): integrate the shape. (ii) given average force and duration: $\bar{F}\tau = \Delta p$. (iii) given F_{max} and shape: $J = F_{max}\tau/k$ with shape factor k (triangle: $J = \frac{1}{2}F_{max}\tau$; half-sine: $J = \frac{2}{\pi}F_{max}\tau$). (iv) given a spring stop: use energy instead; the max force is kx_{max} at max compression.

WHY / the approximation is a singular perturbation: over $\tau \to 0$ the impulse $\int F dt$ stays finite while the displacement $\int v dt$ vanishes - positions freeze, velocities jump. This is the mathematical licence for 'collision at a point'.

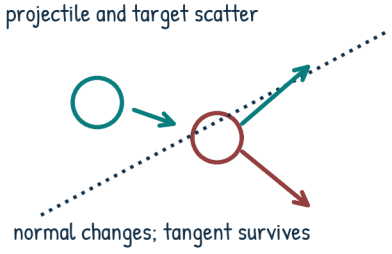
When a finite force acts for a blink: the impulse approximation (co.

CHECK / self-check on OL3 numbers: $J = 2$ N s vs gravity impulse 0.002 N s - ratio 1000; state the ratio in an exam answer to justify the approximation. If the pulse lasted 0.2 s instead of 2 ms, gravity's 0.2 N s would matter - check $mg\tau \ll J$ every time.

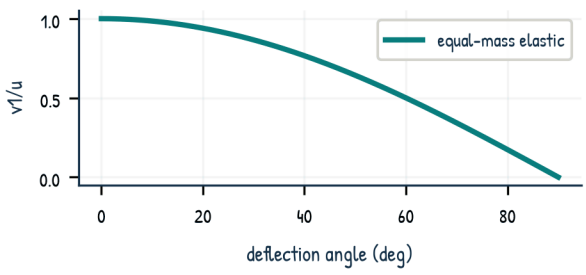
Two balls colliding with a wall simultaneously: each contact delivers its own J; the wall's total impulse is the sum; solve the two pairwise collisions (order irrelevant if contacts are truly simultaneous and independent). A rod on two supports hit at one end:

constraint impulses at BOTH supports enter the momentum ledger - solve with momentum + the rigid-body rotation constraint (advanced; olympiad training).

scatter: labelled physical setup



scatter: limiting curve / diagnostic



REMEMBER / Lab angle: tan(theta) = sin(Theta)/(cos(Theta) + m1/m2); heavy projectiles cannot exceed arcsin(m2/m1).

When a finite force acts for a blink: the impulse approximation

Impulse approximation with scale separation; pulse shapes; multi-body impulses; drills.

Multi-body impulses and constraint impulses (cont.)

Ball hitting a wedge of finite mass: not a wall - use the two-body master formulas along the line of impact (wedge momentum changes; wedge slides). Ball hitting a SPRING: no impulse approximation - solve the SHM matching. The choice of tool is set by the contact duration and the partner's mobility.

CHECK / consistency: for a finite-mass partner, J on the ball and J on the partner are equal/opposite (third law) - if your two J s differ, the solution is wrong.

Impulse drills

(a) 0.2 kg ball at 8 m/s hits a wall and rebounds at 6: $J = 0.2(14) = 2.8$ N s. (b) Same ball dropped 1.25 m on concrete ($e = 0.5$): impact $v = 5$; rebound 2.5; $J = 0.2(7.5) = 1.5$ N s. (c) Triangular pulse peaking at 400 N over 6 ms: $J = \frac{1}{2}(400)(0.006) = 1.2$ N s. (d) Ball 0.15 kg stopped by a 30 N average force: $J = 0.15(25) = 3.75$; $\tau = 3.75/30 = 0.125$ s.

WATCH / Trap: using $J = F_{max}\tau$ for a non-rectangular pulse (overestimates by the shape factor). Trap: forgetting the sign reversal cost $2mv$ vs mv . Trap: comparing impulses of contacts with different durations without normalising to Δp .

Impulse drills (cont.)

CHECK / drill (c) shape factor check: half-sine of same peak and width would give $J = \frac{2}{\pi}(400)(0.006) = 1.53$ N s - shape matters by tens of percent.

When the force history is unknown, bound it: $J = \Delta p$ is exact; if the stop distance d is known, the average force satisfies $\bar{F} = \frac{mv^2}{2d}$ (work-energy) - often more useful than the time-average. Example: 0.15 kg ball at 25 m/s stopped in 0.5 m: $\bar{F} = \frac{mv^2}{2d} = \frac{0.15 \times 625}{2 \times 0.5} = 93.75$ N.

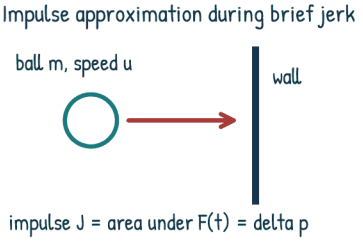
Impulse bounds and force estimates without the pulse shape (cont.)

Compare with a 30 N average force giving $\tau = 0.125$ s (the earlier drill) - two different averages of the same impulse.

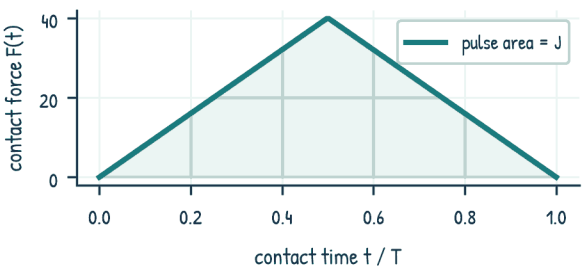
Shape factors: rectangular pulse $J = F_{max}\tau$; triangular $J = \frac{1}{2}F_{max}\tau$; half-sine $J = \frac{2}{\pi}F_{max}\tau$. If a problem quotes only F_{max} , the answer for J must carry the shape assumption explicitly.

CHECK / consistency: the time-average and space-average forms must both reproduce $J = \Delta p$; if your two estimates of J disagree, the assumed shape is inconsistent with the given numbers.

String jerk or bat strike dominates finite gravity during delta t



Peak force scales inversely with contact duration at fixed J



REMEMBER / Impulse approximation: F_impulse dt far exceeds other impulses; gravity is a rounding error in a bat-ball collision.

Ball on a vertically moving platform: the energy-pump machine

Bounce map v' = (1+e)u - ev; lift frame solution; energy pump; drills.

Ball on a vertically moving platform: the energy-pump machine

A platform moves vertically with prescribed speed law $u(t)$; a ball is released above it with restitution e at every contact. One-bounce law (CM-21): up-positive, ball velocity v (negative if falling), platform u: $v_{after} = (1+e)u - ev_{before}$. If the platform RISES at u into a ball falling at speed w: $v_{after} = (1+e)u + ew$ upward.

The platform can therefore pump the ball to arbitrary heights if it keeps rising at the right moments - this is the 'bouncing ball on a piston' machine.

Ball on a vertically moving platform: the energy-pump machine (c.

Two regimes. (1) Moving lift at constant speed U (upward): ball dropped in a lift rising at U: relative problem in the lift frame: ball speed relative w, bounce ew; the LAB speed after n bounces grows like a geometric ladder that converges if $e < 1$ to a fixed relative speed where the per-bounce gain $(1+e)U$ balances the per-bounce loss $(1-e)w$: $w_* = \frac{(1+e)U}{1-e}$ in the platform frame.

Ball on a vertically moving platform: the energy-pump machine (c.

(2) Oscillating platform: resonance when the platform arrives upward exactly as the ball arrives - phase locking; POGO/springboard questions use this.

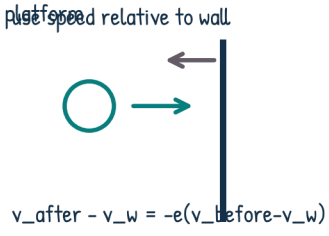
The classic exam configuration (source OL6, with the pure bounce ladder in source OL4): platform initially at rest accelerates up at constant a while a ball bounces with e; find the speed after n bounces or the conditions for the ball to keep bouncing. Method: time between bounces $t_n = 2v_n/g_{rel}$ (with platform acceleration folded into an effective g in the platform frame:

Ball on a vertically moving platform: the energy-pump machine (c.

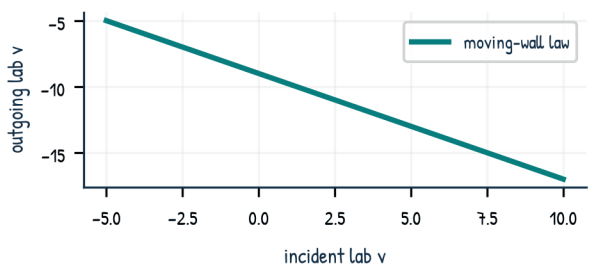
$g_{eff} = g + a$ for a platform accelerating UP); apply the bounce law at each contact; the sequence can be summed if e is constant.

WHY / the machine is the olympiad playground for relative-speed restitution: every contact is a one-line map $v \mapsto (1+e)u - ev$; the composition of maps is where the richness lives. Energy is NOT conserved - the platform's motor does work through the normal force.

platform: labelled physical setup



platform: limiting curve / diagnostic



REMEMBER / Bounce map v_new = (1+e)u - e v; a rising platform pumps energy; in the lift frame it is an ordinary bounce.

Ball on a vertically moving platform: the energy-pump machine (cont.)

Bounce map v' = (1+e)u - ev; lift frame solution; energy pump; drills.

Ball on a vertically moving platform: the energy-pump machine (c.

CHECK / static limit: platform at rest ($u = 0$): $v_{after} = -ev_{before}$; ball on a steadily rising platform in equilibrium: the fixed point of the map $v' = (1+e)U - ev$ is $v = U$ - the ball just rides at the platform speed, bouncing at zero relative speed - physical.

Ball dropped from rest relative to the lift floor; lift rises at U (constant). In the lift frame: ball falls with speed growing as gt ; hits at relative speed $w = \sqrt{2gh}$ (h = drop height in the frame); rebounds at ew ; rises to e^2h ; the cycle repeats with $h \to e^2h$. In the LAB frame each rebound speed is $U + ew$ (from the map with $v_{before} = U - w$):

Worked: ball dropped in a lift rising at constant speed U (cont.)

$v_{after} = (1+e)U - e(U-w) = U + ew$; relative speed up $= ew$ - consistent with the frame picture.

Steady state: the relative speed after bounce is $e^n w_0 \to 0$ for $e < 1$; the ball settles to riding at U with tiny hops. For $e = 1$: relative speed preserved; the ball keeps bouncing forever at the same relative amplitude, drifting with the lift. Energy in the lab: KE plus PE grows as the lift rises - the motor pays.

CHECK / relative-frame audit: every bounce in the lift frame is the textbook fixed-floor bounce with e ; the lab is a pure velocity shift. If a lab-frame calculation disagrees with the frame-shifted answer, the sign convention broke.

Platform-machine drills

(a) Ball dropped 2 m in a lift rising at 3 m/s, $e = 0.5$, $g = 10$: relative impact speed $w = \sqrt{40} = 6.32$; lab after $= 3 + 0.5(6.32) = 6.16$ m/s up. (b) Platform rising at 2 m/s meets ball falling at 4 m/s (lab), $e = 0.8$: $v_{after} = (1.8)(2) + 0.8(4) = 6.8$ m/s up. (c) Driven steady relative speed in a lift rising at $U = 1$ with $e = 0.5$:

$w_* = \frac{1.5}{0.5} = 3$ m/s (for periodic kicks; a pure drop decays to 0 instead).

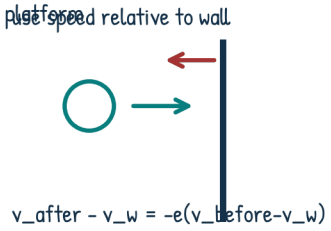
WATCH / Trap: applying the map with lab speeds and forgetting the $(1+e)u$ platform term. Trap: using g instead of $g + a$ in the platform-accelerating case. Trap: assuming energy conservation anywhere in this machine.

Platform-machine drills (cont.)

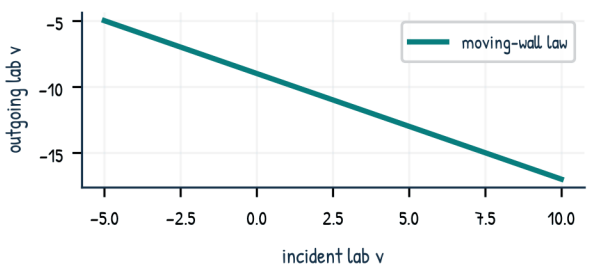
CHECK / drill (b) energy sanity: ball KE grew from $\frac{1}{2}m(16)$ to $\frac{1}{2}m(46)$ - platform motor did work; momentum is NOT conserved (platform is external).

Platform $u(t) = u_0 \cos \omega t$; ball released above. Each bounce maps $v \mapsto (1+e)u(t_n) - ev$ at contact times t_n . Phase locking: if the ball always meets the platform rising near its maximum u_0 , the gain per bounce is maximal $(1+e)u_0 + ew$ - the POGO condition. If the flight time $2v/g$ equals an integer number of platform periods, the phase repeats and the bounce ladder is a clean geometric series.

platform: labelled physical setup



platform: limiting curve / diagnostic



REMEMBER / Bounce map v_new = (1+e)u - e v; a rising platform pumps energy; in the lift frame it is an ordinary bounce.

Ball on a vertically moving platform: the energy-pump machine (cont.)

Bounce map v' = (1+e)u - ev; lift frame solution; energy pump; drills.

Resonance locking on an oscillating platform (cont.)

Numbers: platform $u = 2\cos(2t)$, ball dropped in phase with the peak: first bounce with $w = 5$, $e = 0.5$: in phase: $(1+e)u + ew = 3 + 2.5 = 5.5$ m/s up. Out of phase (platform at $-u_0$): $v_{after} = -3 + 2.5 = -0.5$ - barely bounces.

CHECK / the map's fixed point for constant u is $v = u$ (ride along); for oscillating u there is no fixed point and the amplitude ratchets only when phase-locked - the olympiad question is always about the PHASE.

Chain over a smooth peg: the full solution, including the moment it.

Setup (source OL7): a uniform chain of mass m, length L hangs over a frictionless peg with length x on one side and $L - x$ on the other, released from rest at some $x_0 > L/2$. (1) Acceleration: the unbalanced weight is $\lambda x g - \lambda(L-x)g = \lambda g(2x - L)$ on total mass $m = \lambda L$: $a = \dfrac{g(2x-L)}{L}$. At $x = L/2$: $a = 0$ (unstable equilibrium); at $x = L$: $a = g$ - the whole chain is in FREE FALL at the end (source OL7's key line). (2) Speed: $v\,dv = a\,dx$ gives $v^2 = \frac{2g}{L}\left[(x - \frac{L}{2})^2 - (x_0 - \frac{L}{2})^2\right]$; released from rest exactly at $L/2$ plus a nudge, $v^2 = \frac{2g}{L}(x - \frac{L}{2})^2$.

Chain over a smooth peg: the full solution, including the moment it.

(3) Peg force - the part everyone botches. The peg feels the weight mg MINUS the momentum-flux term of the strand going slack:

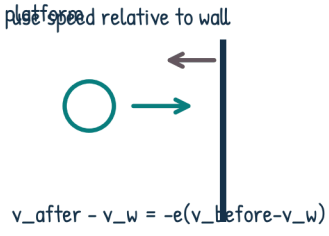
writing the chain's downward momentum $p = \lambda v(2x - L)$ gives $\frac{dp}{dt} = 2\lambda[v^2 + (x-\frac{L}{2})a] = 4\lambda v^2 \,(consistent substitution),\, so\, F_{peg} = mg - 4\lambda v^2 = mg\left[1 - \dfrac{2(2x-L)^2}{L^2}\right]$ for the nudge-from-centre release.

Chain over a smooth peg: the full solution, including the moment it.

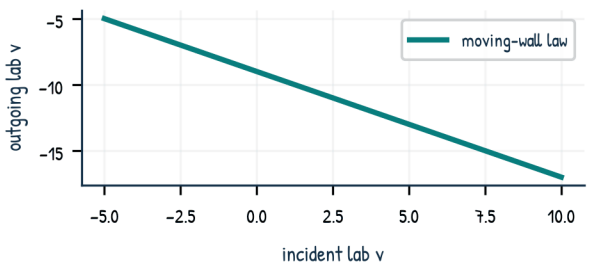
The peg force DECREASES as the chain accelerates - it does not spike. Contact is lost when $F = 0$: $(2x - L)^2 = L^2/2$, i.e. $x = \frac{L}{2}\left(1 + \frac{1}{\sqrt{2}}\right) \approx 0.854L$; beyond that the chain flies off the peg (formal $F < 0$ is unphysical for a peg that can only push).

Why the folklore '3mg at the end' is wrong for THIS problem: the classic 3mg belongs to the chain LANDED ON A SCALE (OLY-01: weight + landing flux both grow). The sliding peg problem has the opposite structure: the moving chain goes slack against the peg as $a \to g$; the end-of-motion peg force tends to zero, not to $3mg$. The two chain problems bracket the flux law:

platform: labelled physical setup



platform: limiting curve / diagnostic



REMEMBER / Bounce map v_new = (1+e)u - e v; a rising platform pumps energy; in the lift frame it is an ordinary bounce.

Chain over a smooth peg: the full solution, including the moment it lets go

Peg solution: a = g(2x-L)/L, v^2(x), F_peg = mg[1-2(2x-L)^2/L^2], contact loss at 0.854L; corrected folklore.

Chain over a smooth peg: the full solution, including the moment it.

onto a pan the flux ADDS to the reading (λv^2 with v of landing); over a peg the flux of the TURNING strand is fed by the chain's own weight and the peg sheds load.

WHY / the momentum-ledger method is the only safe route for moving-chain problems: write $p(t)$ for the whole chain, differentiate, equate to $W - F$. Strand-by-strand tension bookkeeping easily double-counts the flux. The source's ODE method ($v\,dv = a\,dx$) and the momentum method are the two halves of the complete solution.

Chain over a smooth peg: the full solution, including the moment it.

CHECK / limits: $x = L/2$: $F = mg$, $v = 0$ (static). $x \to L$: $a \to g$, $F \to -mg$ (already detached). Energy: $\frac{1}{2}mv^2 =$ PE drop: PE $= -\frac{\lambda g}{2}[x^2 + (L-x)^2]$; drop from $L/2$ to x : $\Delta PE = -\lambda g(x - \frac{L}{2})^2$; KE $= \lambda g(x - \frac{L}{2})^2 = \frac{1}{2}mv^2$ with $v^2 = \frac{2g}{L}(x - \frac{L}{2})^2$ - exactly consistent.

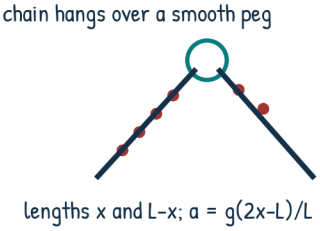
Worked numbers and the released-from-x0 variant

Chain $m = 1$ kg, $L = 1$ m, $g = 10$, released with $x_0 = 0.75$ m (so $x_0 - L/2 = 0.25$): at $x = 0.9$ m: $v^2 = 20[(0.4)^2 - (0.25)^2] = 20(0.0975) = 1.95$: $v = 1.40$ m/s; $a = 10(2(0.9) - 1) = 8$ m/s^2 ; $F = 10[1 - 2(0.8)^2] = -2.8$ N - already detached (contact lost at $x = 0.5(1 + 0.707) = 0.854$ m, before 0.9). At $x = 0.8$ m: $v^2 = 20[(0.3)^2 - (0.25)^2] = 20(0.0275) = 0.55$: $v = 0.74$ m/s; $F = 10[1 - 2(0.6)^2] = 2.8$ N - still in contact.

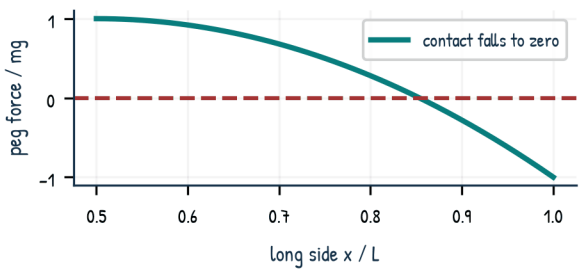
Worked numbers and the released-from-x0 variant (cont.)

Comparison set for exams: (i) chain onto a pan (OLY-01): reading $3\lambda xg$ to $3mg$ at the end - flux ADDS; (ii) chain over a peg (this sheet): peg force $mg[1 - 2(2x - L)^2/L^2]$, contact loss near the end - flux of the turn is INTERNAL once the chain free-falls; (iii) chain LIFTED at constant speed (CM-18): $\lambda xg + \lambda v^2$ - both terms prescribed.

chain over peg: unequal sides x and L-x



peg: limiting curve / diagnostic



REMEMBER / Chain over a peg: a = g(2x-L)/L, v^2 = (2g/L)(x - L/2)^2; peg force falls to zero at x = 0.854L - the chain lets go.

Chain over a smooth peg: the full solution, including the moment it lets go (cont.)

Peg solution: a = g(2x-L)/L, v^2(x), F_peg = mg[1-2(2x-L)^2/L^2], contact loss at 0.854L; corrected folklore.

Worked numbers and the released-from-x0 variant (cont.)

The diagram of what flows where decides which of the three structures applies.

CHECK / the detachment fraction is geometry-independent of m, g: $x_{detach}/L = \frac{1}{2}(1+1/\sqrt{2}) = 0.854$ exactly - a pure number worth quoting in an exam answer as the 'loss of contact' condition.

Worked numbers and the released-from-x0 variant (cont.)

Toolkit: PE of hanging strand length ℓ is $-\frac{\lambda g \ell^2}{2}$ (COM at half depth); energy gives $v^2(x)$; $p = \lambda v(\ell_1 - \ell_2)$ gives the ledger; $F_{peg} = W - dp/dt$; the turn contributes a flux $2\lambda v^2$ per strand junction when the chain CONTINUES over the peg (e.g. chain poured over a peg into a bucket) - the sign of the flux term depends on whether the strand downstream is pulling or slack.

Peg-problem toolkit and drills (cont.)

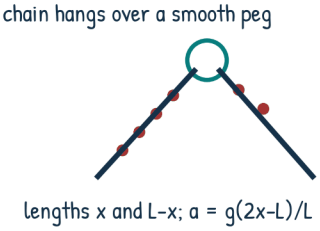
Drill (a): chain released with $x_0 = 0.6L$: at $x = 0.75L$: $v^2 = \frac{2g}{L}[(0.25L)^2 - (0.1L)^2] = \frac{2g}{L}(0.0525L^2) = 0.105gL$; $F = mg[1 - 2(0.5)^2] = 0.5mg$. (b) At what x is $v = \sqrt{gL/2}$? $v^2 = \frac{2g}{L}(x - \frac{L}{2})^2 = \frac{gL}{2}$: $(x - \frac{L}{2})^2 = \frac{L^2}{4}$:

Peg-problem toolkit and drills (cont.)

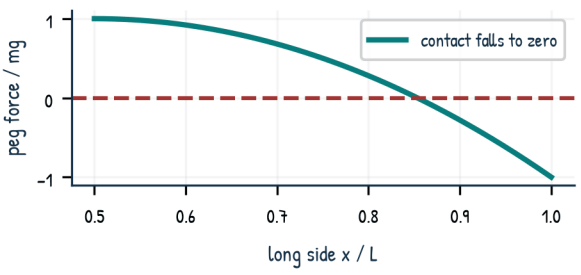
$x = L$ - only at the very end (for nudge release). (c) Show the chain reaches $a = g$ exactly when it would leave the peg: $F = mg - 4\lambda v^2$ and $4\lambda v^2 = mg$ at x_{detach}; the free-fall line is approached as contact fades.

for a chain FLOWING over a peg into motion on both sides (no slack): $F = W + 2\lambda v^2$ (flux ADDS) - e.g. the fold-release family of OLY-08. Slack vs pulling downstream is the single most important qualitative judgement in chain problems.

chain over peg: unequal sides x and L-x



peg: limiting curve / diagnostic



REMEMBER / Chain over a peg: a = g(2x-L)/L, v^2 = (2g/L)(x - L/2)^2; peg force falls to zero at x = 0.854L - the chain lets go.

Chain over a smooth peg: the full solution, including the moment it lets go (cont.)

Peg solution: a = g(2x-L)/L, v^2(x), F_peg = mg[1-2(2x-L)^2/L^2], contact loss at 0.854L; corrected folklore.

Peg-problem toolkit and drills (cont.)

CHECK / drill (a): $0.25^2 - 0.1^2 = 0.0525$; $F = 0.5mg$.

Tension at a point of the moving chain: cut the chain; $T =$ (weight of the part beyond the cut) + (flux of momentum across the cut) + (mass beyond x acceleration). For the peg problem: the strand tension just below the peg is $T = \lambda g\ell + \lambda\ell a$ for a hanging strand of length ℓ accelerating at a (the strand's own weight plus its acceleration). At the nudge-release with $\ell = L - x$: $T = \lambda g(L - x) + \lambda(L - x)a$.

Tension along the moving chain (cont.)

Drill: $L = 1$, $x = 0.7$ (nudge release): $a = 2g(0.2) = 4$; $\ell = 0.3$: $T = 0.3(10) + 0.3(4) = 4.2$ N for $\lambda = 1$ kg/m. The peg force is the sum of the two strand tensions at the peg (which equals $mg - 4\lambda v^2$ as derived) - the two bookkeepings agree.

CHECK / at $x = L/2$ (equilibrium, $v = 0$, $a = 0$): $T = \lambda gL/2 = mg/2$ each side; the peg holds mg total.

Why rockets have stages: drop the dead metal

A rocket stage burns its propellant then throws away its empty tank and engine (structural mass). The next stage starts from a lighter vehicle. Model (source OL8): total mass M, payload M_p; structural coefficient $\epsilon = m_{structure}/m_{initial\ of\ stage}$; exhaust speed v_e. A stage with initial mass m_i burning until m_f gives $\Delta v = v_e \ln(m_i/m_f)$;

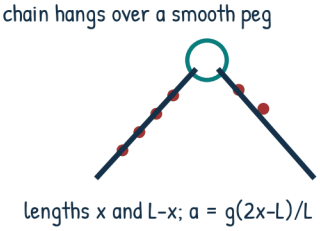
at burnout the empty structure ϵm_i is DISCARDED before the next burn. That discard is what breaks the telescoping of the logs - and it is the entire benefit of staging.

Why rockets have stages: drop the dead metal (cont.)

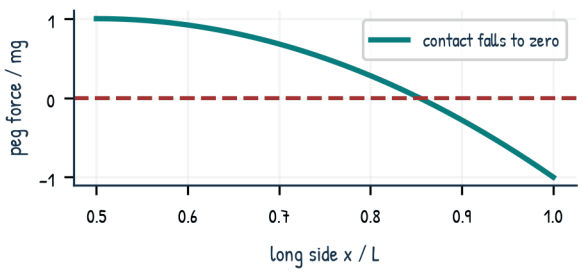
Mass timeline discipline: draw $M_0 \to$ (burn 1) $\to m_1 \to$ (discard structure) $\to m_2 \to$ (burn 2) $\to m_f$. Then $\Delta v = v_e[\ln(M_0/m_1) + \ln(m_2/m_f)]$ with $m_2 = m_1 - m_{structure,1}$. Forget the discard and the two logs telescope to the useless $v_e \ln(M_0/m_f)$; discard BEFORE burnout and you threw away propellant mass you had not used.

Numbers (source P30 with consistent bookkeeping; the source's printed staging mixes two split conventions - recorded in SOURCE-AUDIT): $M_0 = 1000$ kg, two stages of 400 kg propellant + 50 kg structure each, payload $M_p = 100$ kg: stage 1 burns to 600 kg, discard 50, stage 2 starts at 550, burns to 150:

chain over peg: unequal sides x and L-x



peg: limiting curve / diagnostic



REMEMBER / Chain over a peg: a = g(2x-L)/L, v^2 = (2g/L)(x - L/2)^2; peg force falls to zero at x = 0.854L - the chain lets go.

Why rockets have stages: drop the dead metal

Staging ledger; mass timeline; corrected P30 numbers (5.43 vs 4.83 km/s); drills.

Why rockets have stages: drop the dead metal (cont.)

$\Delta v = v_e[\ln(1000/600) + \ln(550/150)] = v_e(0.511 + 1.299) = 1.81 v_e = 5.43$ km/s at $v_e = 3$ km/s. Single-stage with the same total propellant and structure (100 kg kept to the end): $\Delta v = 3\ln(1000/200) = 3\ln 5 = 4.83$ km/s. Staging wins by about 12% with identical hardware.

Why rockets have stages: drop the dead metal (cont.)

WHY / the rocket equation punishes carried mass logarithmically; dropping empty tanks deepens the mass ratio per burn. With zero structural mass any split gives the same $\Delta v = v_e \ln(M/M_p)$ - structure is the ONLY reason to stage.

CHECK / limits: $\epsilon \to 0$: staged = unstaged; $M_p \to M$: $\Delta v \to 0$; more stages (fixed hardware) always increase Δv until the interstage hardware itself dominates. Timeline closes: $400 + 400 + 50 + 50 + 100 = 1000$ kg.

Optimal split and the multistage ladder

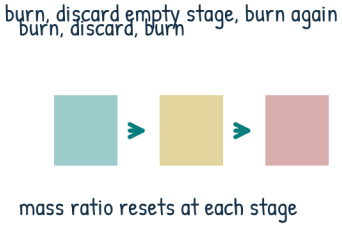
For two identical stages (equal ϵ , equal v_e): the total $\Delta v = v_e[\ln(M/m_1) + \ln(m_2/m_f)]$ under the mass-timeline constraint; the optimisation (Jensen on the logs) equalises the two stage Δv s for maximum total at fixed hardware. With n stages and fixed total structure mass, more stages keep helping until interstage hardware eats the gain - real rockets stop at $n = 2$ or 3.

Optimal split and the multistage ladder (cont.)

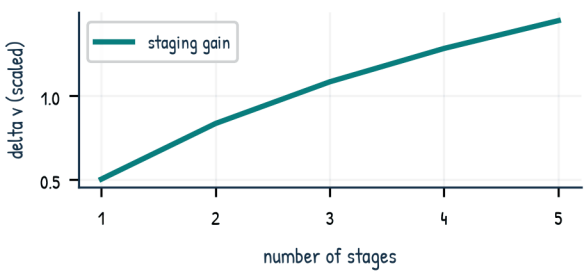
With gravity: $\Delta v_{net} = \sum v_e \ln(m_i/m_f) - g t_{burn}$; staging also keeps acceleration high in thin air, cutting gravity loss - the engineering trade-off beyond the textbook ledger. The same ledger governs 'drop ballast mid-journey' puzzles and two-block spring mass-shedding (CM-14).

for identical stages the optimal equal-split principle is a one-line AM-GM argument: $\ln a + \ln b = \ln(ab)$ is maximal at fixed $a + b$ when $a = b$ - equal logs, equal Δv .

staging: labelled physical setup



staging: limiting curve / diagnostic



REMEMBER / Staging: with zero structure any split gives v_e ln(M/M_p); the real gain is dropping spent tanks.

Why rockets have stages: drop the dead metal (cont.)

Staging ledger; mass timeline; corrected P30 numbers (5.43 vs 4.83 km/s); drills.

Optimal split and the multistage ladder (cont.)

CHECK / single vs staged audit in ratio form: $\frac{\Delta v_{staged}}{\Delta v_{single}} = \frac{\ln(1000/600)+\ln(550/150)}{\ln 5} = \frac{1.810}{1.609} = 1.125$ - exactly 12.5% more Δv.

(a) Single stage: $M = 1000$, propellant 800, structure 100, payload 100: $\Delta v = v_e \ln 5 = 1.61 v_e$. (b) Two stages (the worked case): $1.81 v_e$ - beats (a) by $0.20 v_e$. (c) Three stages of 266.7 propellant + 33.3 structure each, payload 100: the timeline is $1000 \to 733.3 \to 700 \to 433.3 \to 400 \to 133.3$ (burn, discard, burn, discard, burn;

Staging drills (cont.)

the last structure rides along). Then $\Delta v = v_e[\ln(1000/733.3)+\ln(700/433.3)+\ln(400/133.3)] = v_e[0.310+0.480+1.099] = 1.89 v_e$ - three stages beat two, which beat one: monotone improvement.

WATCH / Trap: discarding before burnout; telescoping by forgetting the discard; mixing 'structure of stage' with 'payload'. The mass timeline diagram is the answer sheet.

CHECK / drill (c) timeline: $266.7 \times 3 + 33.3 \times 3 + 100 = 800 + 100 + 100 = 1000$; final $133.3 = 100 + 33.3$ (last structure rides along).

Inverse staging: sizing the rocket for a mission

Given Δv target and v_e: required mass ratio $m_0/m_f = e^{\Delta v/v_e}$. Mission $\Delta v = 9$ km/s at $v_e = 3$ km/s: ratio $= e^3 = 20.1$ - the payload + structure must be under 5% of launch mass. With structural coefficient ϵ per stage, n stages give successively deeper ratios; the practical answer: two stages at $\epsilon = 0.1$ reach ratios around 10-15 - matching real launchers.

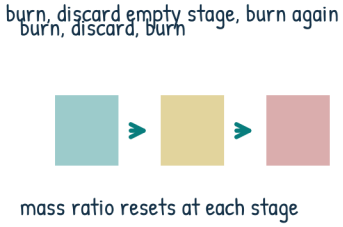
Gravity-loss back-of-envelope: at thrust-to-weight 1.3, the gravity loss $\approx g t_{burn}$ quoted from the burn time, not from Δv alone; a 9 km/s mission at $a = 2g$ loses roughly half a km/s to gravity in the first minutes.

Inverse staging: sizing the rocket for a mission (cont.)

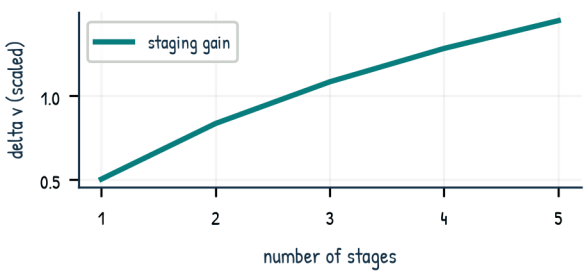
CHECK / reverse the worked case: 5.43 km/s at $v_e = 3$ needs ratio $e^{1.81} = 6.1$; the staged design delivers $1000/150 = 6.67$ - consistent with the timeline (final mass includes the last structure).

A circular jet of area A, speed v, density ρ strikes a large fixed flat plate and spreads along it (source OL9). Mass rate $\dot{m} = \rho A v$; the normal momentum is destroyed (the water leaves tangentially): $F = \dot{m} v = \rho A v^2$. For a plate tilted at angle θ to the jet axis (normal force only): $F_n = \rho A v^2 \cos\theta$; at angle θ to the PLATE surface:

staging: labelled physical setup



staging: limiting curve / diagnostic



REMEMBER / Staging: with zero structure any split gives v_e ln(M/M_p); the real gain is dropping spent tanks.

Jet on a plate: F = rho A v^2, and the moving-plate power game

Jet flux law; moving plate P = rho A (v-u)^2 u with u = v/3 optimum; cup factor 2; drills.

Jet on a plate: F = rho A v^2 & power game (cont.)

$F_n = \rho A v^2 \sin\theta$ - read the angle definition from the diagram before multiplying.

Moving plate at speed u in the jet's direction: the relative speed is $v-u$, so $F=\rho A(v-u)^2$ and the power delivered to the plate is $P=Fu=\rho A(v-u)^2 u$.

Maximise over u: $dP/du=0$ gives $u=v/3$ (not $v/2$!) with $P_{max}=\frac{4}{27}\rho A v^3$. The turbine designer's law: the jet should be slowed to one third. At $u=0$ (fixed plate) $P=0$;

at $u=v$ (plate flees with the jet) $F=0$ and $P=0$; the maximum sits between.

Jet on a plate: F = rho A v^2 & power game (cont.)

If the water does not spread but rebounds elastically (curved cup): the normal momentum REVERSES: $F=2\dot{m}v=2\rho A v^2$ (the cup variant of source OL9) - double the flat-plate force. A Pelton-wheel cup nearly doubles the force of a flat target; the same factor-2 logic as the turn flux in chain problems.

WHY / the jet is the cleanest possible flux problem: control volume = the water in contact; momentum in $\rho A v^2$ per second along the normal; momentum out zero (flat) or reversed (cup). If the jet discharges to atmosphere there are no pressure terms - the whole force is momentum flux.

Jet on a plate: F = rho A v^2 & power game (cont.)

CHECK / numbers: $A=6$ cm^2, $v=12$ m/s, $\rho=1000$: $\dot{m}=7.2$ kg/s; $F=86.4$ N; on a cup: 172.8 N; at $u=4$ m/s: $F=1000(6\times10^{-4})(64)=38.4$ N, $P=153.6$ W.

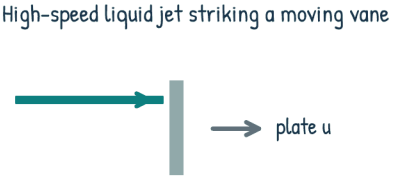
A hose or gun ejecting $\dot{m}$ at v_e feels recoil $\dot{m}v_e$ - the rocket thrust (CM-17) in static form. A fire hose of 5 kg/s at 20 m/s: 100 N on the firefighter. A sprinkler cart: thrust = mass rate x relative jet speed; the cart accelerates like a rocket with water as exhaust. The jet-pump / turbine family: same flux law with the right control volume.

Recoil as flux; the jet taxonomy (cont.)

Water falling on a moving horizontal belt: retarding force $=\dot{m}(v-u)$ with u the belt speed (CM-16 family). The exam taxonomy: jet on plate $\rho A v^2$; cup $2\rho A v^2$; moving plate $\rho A(v-u)^2$; belt pickup $\dot{m}v$; rocket $\dot{m}v_e$. One law, five costumes.

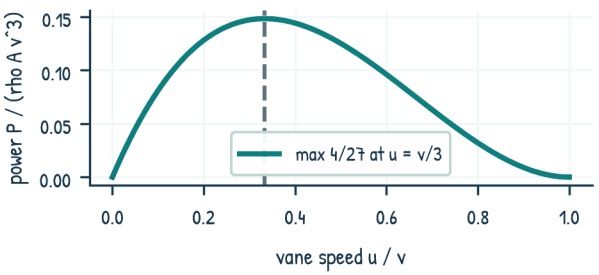
CHECK / recoil audit: 0.5 kg/s at 15 m/s: 7.5 N; pump power $=\frac{1}{2}\dot{m}v^2=56$ W = the jet's KE rate; the plate absorbs Fu on top of that in the belt-split sense - consistent with CM-16's factor-2 bookkeeping.

Moving vane intercepts mass rate rho A (v - u), not rho A v



F = rho A (v - u)^2, peak power at u = v/3

Mechanical power extracted by flat vane peaks at u = v / 3



REMEMBER / Jet on a plate: F = rho A v^2 normal component; on a moving plate the relative speed (v - u) does the work.

Jet on a plate: F = rho A v^2, and the moving-plate power game (cont.)

Jet flux law; moving plate P = rho A (v-u)^2 u with u = v/3 optimum; cup factor 2; drills.

Jet drills

(a) Jet $A = 4$ cm^2, $v = 10$ m/s on a flat plate: $F = 40$ N. (b) Same on a plate at 45 degrees to the jet axis (normal force): $F_n = 28.3$ N. (c) Moving plate at 4 m/s: $F = 1000(4 \times 10^{-4})(36) = 14.4$ N; $P = 57.6$ W. (d) Cup reversal: double (a): 80 N. (e) Optimal turbine speed for a 20 m/s jet: $u = 6.67$ m/s; $P_{max} = \frac{4}{27}(1000)(4 \times 10^{-4})(8000) = 474$ W.

Jet drills (cont.)

WATCH / Trap: using v instead of $v - u$ for moving plates. Trap: angle measured from the wrong reference. Trap: missing the factor 2 for rebound cups. Trap: equating P_{max} with the jet's KE rate $\frac{1}{2}\rho A v^3$ - the best useful power is only $\frac{8}{27}$ of the jet's KE rate, about 30%.

CHECK / drill (e) consistency: $u = v/3 = 6.67$; $F = \rho A(2v/3)^2 = 1000(4 \times 10^{-4})(177.8) = 71.1$ N; $P = 71.1 \times 6.67 = 474$ W.

Efficiency curve of a turbine stage

Useful power $P = \rho A(v-u)^2 u$ rises from 0 at $u = 0$, peaks at $u = v/3$ with $P_{max} = \frac{4}{27}\rho A v^3$, falls to 0 at $u = v$. Efficiency $\eta = P/(\frac{1}{2}\rho A v^3)$: at the peak $\eta = 8/27 = 30$ %. The Pelton cup shifts the force constant up by the factor 2 in the flux; the $u = v/3$ optimum is unchanged.

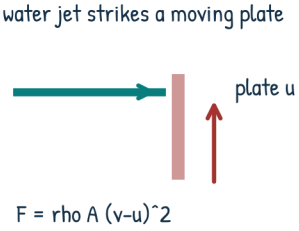
Drill: jet $v = 15$ m/s, $\rho A = 0.5$ kg/m: $P_{max} = \frac{4}{27}(0.5)(3375) = 250$ W at $u = 5$ m/s. Flat plate at $u = 5$: $F = 0.5(100) = 50$ N, $P = 250$ W.

Efficiency curve of a turbine stage (cont.)

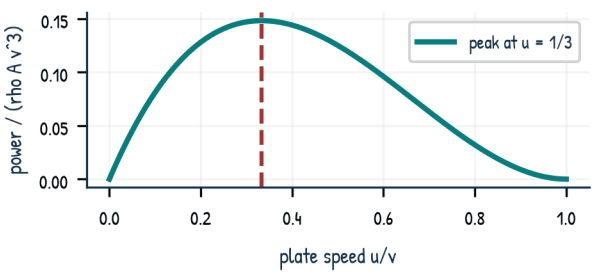
CHECK / curve sanity: $P(v/4) = \rho A(3v/4)^2(v/4) = \frac{9}{64}\rho A v^3 = 0.141$ vs peak 0.148 - near-flat top; real impulse turbines cluster u/v near 1/2 with cups.

Clean statement (source OL10, whose 'COM \L/4' phrasing is ambiguous - recorded in SOURCE-AUDIT; this is the standardised version): a uniform chain of mass m, length L is folded at its MIDPOINT and laid on a smooth horizontal table with the fold just past the smooth edge, so BOTH halves hang together. Released from rest.

jet: labelled physical setup



jet: limiting curve / diagnostic



REMEMBER / Jet on a plate: F = rho A v^2 normal component; on a moving plate the relative speed (v - u) does the work.

Folded chain released over a smooth edge: v = y sqrt(2g/L)

Folded chain: v = y sqrt(2g/L), departure sqrt(gL/2); ambiguity guard for OL10; drills.

Folded chain released over a smooth edge: v = y sqrt(2g/L) (cont.)

Let y = the descent of the fold = the length of EACH hanging strand (total hanging length $2y$). Speed: energy with hanging mass $2\lambda y$ and COM at depth $y/2$: PE loss $= 2\lambda y \cdot g \cdot \frac{y}{2} = \lambda g y^2$; KE $= \frac{1}{2}\lambda L v^2$ (the WHOLE chain moves at speed v - the fold is a rigid corner). So $\frac{1}{2}\lambda L v^2 = \lambda g y^2$:

$v = y\sqrt{\frac{2g}{L}}$ - speed linear in descent, NOT $\sqrt{2gy}$ as for a free particle.

Folded chain released over a smooth edge: v = y sqrt(2g/L) (cont.)

Departure: the chain leaves the table when the fold has descended $y = L/2$ (both strands fully hanging): $v_{dep} = \frac{L}{2}\sqrt{\frac{2g}{L}} = \sqrt{\frac{gL}{2}}$. After that the chain falls free with its departure speed. The law $v \propto y$ means constant acceleration:

$a = \frac{dv}{dt} = \sqrt{\frac{2g}{L}}\, v = \frac{2g}{L} y$ - the fold accelerates in proportion to its own depth (a linear instability, kin to the chain-over-peg ODE of OLY-05).

Folded chain released over a smooth edge: v = y sqrt(2g/L) (cont.)

Force on the edge (flux bonus): the doubled strand turns over the edge at rate $2\lambda v$; the edge feels hanging weight plus a turn-flux term of the $2\lambda v^2$ family with the doubled density. The exam answer quotes the energy line and $v_{dep} = \sqrt{gL/2}$; the force bookkeeping is a fair exercise on the momentum ledger of OLY-05.

WHY / the fold constrains BOTH halves to move together - one velocity, one degree of freedom. That is why the whole chain's KE uses L while the PE uses only the hanging $2y$: the energy equation is unusually clean and $v = y\sqrt{2g/L}$ is exact.

Folded chain released over a smooth edge: v = y sqrt(2g/L) (cont.)

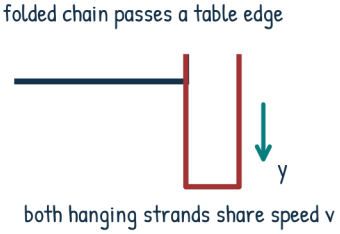
CHECK / limits: $y = 0$: $v = 0$; $y = L/2$: $v = \sqrt{gL/2}$; $g \to 0$: $v \to 0$. Units: $y\sqrt{g/L} =$ m $\times$ s^{-1} = m/s.

Compare free fall through y: $\sqrt{2gy}$ vs fold $y\sqrt{2g/L}$ - at $y = L/2$: free $\sqrt{gL}$ vs fold $\sqrt{gL/2}$: the fold is SLOWER (half the chain is still on the table).

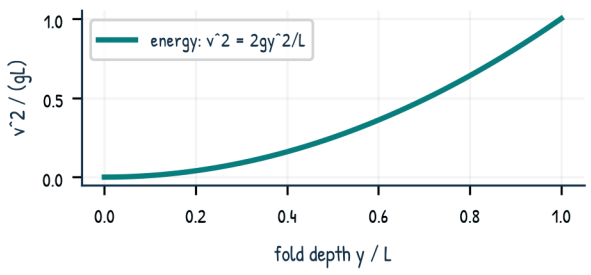
Variant A: fold NOT at the midpoint (strands of unequal length): the shorter strand runs out first; beyond that it is a single-hanging-strand problem - piecewise solve.

Variant B: fold on the table with the fold-point LIFTED then released (the 'COM + L/4' reading of source OL10): the answer depends on the exact initial shape - define the geometry in the diagram before solving.

fold at edge: equal hanging strands, descent y



fold: limiting curve / diagnostic



REMEMBER / Folded chain off a smooth edge: v = y sqrt(2g/L) with y the fold depth; leaves the table at sqrt(gL/2).

Folded chain released over a smooth edge: v = y sqrt(2g/L) (cont.)

Folded chain: v = y sqrt(2g/L), departure sqrt(gL/2); ambiguity guard for OL10; drills.

Variant geometries and the ambiguity guard (cont.)

Variant C: rough table: friction on the table-lying part retards; energy loses the friction-work term. Variant D: the chain is pushed initially: from $\frac{1}{2}\lambda L v^2 = \frac{1}{2}\lambda L v_0^2 + \lambda g y^2$: $v^2 = v_0^2 + \frac{2gy^2}{L}$.

CHECK / variant D limit: $y = 0$ gives $v = v_0$; at $y = L/2$, $v_0 = 0$ recovers $v = \sqrt{gL/2}$.

(a) $L = 2$ m, $g = 10$: $v = y\sqrt{10} = 3.16y$; at $y = 0.4$ m: $v = 1.26$ m/s; departure at $y = 1$ m: $v = 3.16$ m/s ($= \sqrt{gL/2} = \sqrt{10}$). (b) Unequal fold (strands $L/4$ and $3L/4$ hanging, i.e. NOT a doubled fold): different problem - use the peg-style ODE of OLY-05. (c) Compare departure speeds: fold $\sqrt{gL/2}$;

Fold drills (cont.)

single strand sliding off a peg to $x = L$: $v = \sqrt{gL/2}$ as well - the two ODEs are first cousins.

WATCH / Trap: treating the fold as two independent strands (they are kinematically tied - ONE v). Trap: using $\sqrt{2gy}$ (free fall) for the fold. Trap: forgetting the on-table part of the chain in the KE sum.

CHECK / drill (a): $\sqrt{2g/L} = \sqrt{10} = 3.16$; $v(0.4) = 1.26$ m/s; departure $\sqrt{10} = 3.16$ m/s.

Fold as a variable-mass Atwood

The fold's equation $a = \frac{2g}{L}y$ is a variable-mass Atwood: the driving weight $2\lambda yg$ acts on the accelerating mass λL: $a = \frac{2\lambda yg}{\lambda L} = \frac{2gy}{L}$ - the same result as the energy route. The Atwood perspective shows WHY the fold accelerates: the hanging weight grows while the inertia stays fixed.

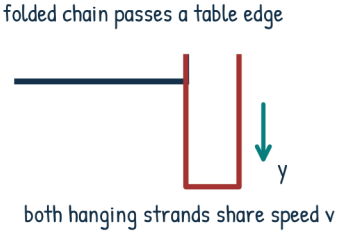
Compare: classic Atwood (two fixed masses): $a = g\frac{m_1 - m_2}{m_1 + m_2}$ = const; fold: $a \propto y$ = exponential runaway; peg: $a = g(2x - L)/L$ = linear. Three ODE shapes in one family: constant, linear, linear-in-y - each with its own $v^2(x)$.

Fold as a variable-mass Atwood (cont.)

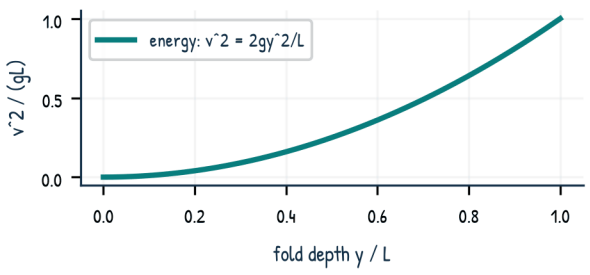
CHECK / Atwood limit of the fold: freeze y at y_0 (clamp): $a = 2gy_0/L$ constant - reduces to a fixed-mass Atwood with effective masses $2\lambda y_0$; the clamp force appears - consistent bookkeeping.

The demand: conservation of momentum must hold in ALL inertial frames. If $p = mv$ with invariant m, a Lorentz transformation of a simple collision gives different momenta in different frames - the law fails. The unique form (up to an overall constant) that transforms correctly is $\mathbf{p} = \gamma m \mathbf{v}$ with $\gamma = 1/\sqrt{1 - v^2/c^2}$.

fold at edge: equal hanging strands, descent y



fold: limiting curve / diagnostic



REMEMBER / Folded chain off a smooth edge: v = y sqrt(2g/L) with y the fold depth; leaves the table at sqrt(gL/2).

Why momentum must be gamma m v (source OL5)

p = gamma m v from frame-independence; scope fence for classical mechanics; preview of PART 28.

Why momentum must be gamma m v (source OL5) (cont.)

Derivation sketch (the standard two-frame elastic collision argument): write $p = mf(v)$; demand that a collision viewed in the lab and in a boosted frame both conserve p; the functional equation forces $f = \gamma$.

Consequences previewed here (full treatment is relativistic dynamics): (1) $\gamma \approx 1 + \frac{v^2}{2c^2}$ at low speed, so $p \approx mv$ up to tiny corrections - Newtonian mechanics is the $v \ll c$ shadow. (2) Force is dp/dt not ma: $F = \gamma^3 ma$ along the motion - a constant force cannot accelerate a mass to c.

Why momentum must be gamma m v (source OL5) (cont.)

(3) Relativistic KE $= (\gamma - 1)mc^2 \to \frac{1}{2}mv^2$ as $v \to 0$; the full series is $\frac{1}{2}mv^2 + \frac{3}{8}\frac{mv^4}{c^2} + \dots$ (4) The 'mass increase' $m \to \gamma m$ is a historical way of writing the same formula - the modern view keeps m invariant and changes the momentum law (source OL5's Significance).

For classical olympiad/JEE mechanics: the exam-relevant point is the scope statement - all our conservation laws assume $v \ll c$; when speeds approach c, momentum conservation still holds but with $p = \gamma mv$ and e-type restitution becomes frame-dependent and subtle. Keep this sheet as the fence around the classical playground.

Why momentum must be gamma m v (source OL5) (cont.)

WHY / the deep principle: conservation laws are statements about symmetry (translation invariance of space and time). The relativistic momentum is what survives when both symmetries act together - the classical mv is the low-speed limit of the correct object, not a definition.

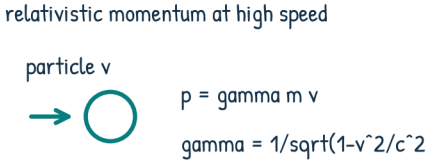
CHECK / low-speed limit $v/c = 0.1$: $\gamma = 1.005$: a 0.5% correction to p - invisible in every chapter problem. $v = 0.9c$: $\gamma = 2.29$: more than double mv - classical mechanics gone.

The two-frame argument in one paragraph (cont.)

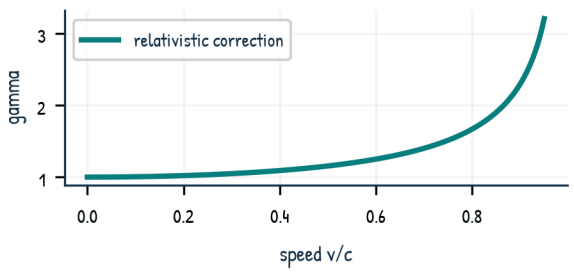
Take an elastic collision between identical particles, symmetric in the CM. Boost along the collision axis. In Newtonian physics the velocity addition $u' = u - V$ preserves momentum conservation. In relativity, velocity addition is $u' = \frac{u - V}{1 - uV/c^2}$;

requiring conservation in both frames with $p = mf(u)$ gives a functional equation for f whose unique regular solution is $f(u) = \gamma u$ (up to constant m).

relativ: labelled physical setup



relativ: limiting curve / diagnostic



REMEMBER / p = gamma m v is the unique momentum that conservation keeps covariant; mass stays invariant.

Why momentum must be gamma m v (source OL5) (cont.)

p = gamma m v from frame-independence; scope fence for classical mechanics; preview of PART 28.

The two-frame argument in one paragraph (cont.)

Rest mass is invariant: every observer agrees on m; they disagree on p and E. Energy-momentum forms a four-vector $(E/c, \mathbf{p})$; $E^2 = (pc)^2 + (mc^2)^2$ unifies the ledger: massless particles have $p = E/c$ (photons); massive at rest have $E = mc^2$.

relativistic collisions still use the same triage: choose the frame (CM is best), conserve the four-momentum, transform back. The 90-degree rule of CM-12 becomes 'the angle opens ABOVE 90 for relativistic equal-mass collisions in the lab' - a classic modern-physics exam twist.

The two-frame argument in one paragraph (cont.)

CHECK / energy-momentum checks: photon $m = 0$: $E = pc$; electron at rest: $p = 0$, $E = mc^2$; Newtonian ledger recovered for $E - mc^2 \ll mc^2$: $E \approx mc^2 + \frac{p^2}{2m}$.

(1) Momentum conservation is the more fundamental law; $F = ma$ is its low-speed shadow. (2) The classical chapter's formulas (3.7, 4.5, 4.6) are all $v \ll c$ statements - state the assumption when speeds are given as 'large'. (3) The rocket equation assumes non-relativistic exhaust; relativistic rockets integrate $\Delta p = v_e dm$ with γ folded in.

What to carry back into classical mechanics (cont.)

(4) Restitution (e) has no clean relativistic definition for extended bodies - modern physics avoids it.

Exam context: this sheet is the olympiad extension layer (the source lists it as OL5, a PART-28 preview). For JEE the one-line takeaway: $p = \gamma mv$ reduces to mv at low speed and momentum is conserved in all frames - the syllabus does not require the derivations.

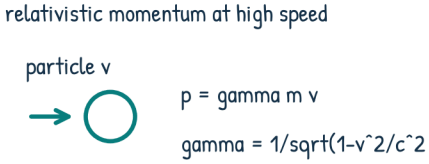
CHECK / limit drill: $\gamma - 1 \approx v^2/2c^2$ at $v = 0.01c$: 5×10^{-5} - the classical error is one part in 20,000.

Gamma drills

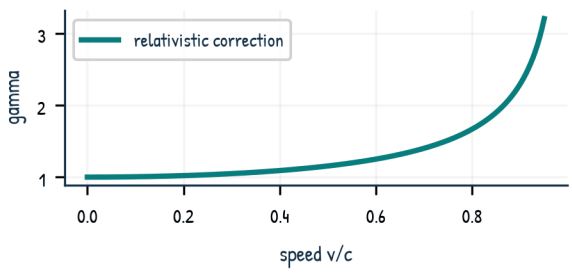
(a) $v = 0.5c$: $\gamma = 1.155$; (b) $v = 0.8c$: $\gamma = 1.667$; (c) $v = 0.99c$: $\gamma = 7.09$; (d) KE at 0.8c $= (\gamma - 1)mc^2 = 0.667mc^2$ - twice the Newtonian $\frac{1}{2}m(0.64c^2) = 0.32mc^2$. (e) A proton at rest has $E = 938$ MeV; at $\gamma = 4$: total 3.75 GeV, KE 2.81 GeV.

Momentum at 0.8c for 1 kg: $p = \gamma mv = 1.667(1)(2.4 \times 10^8) = 4 \times 10^8$ kg m/s vs Newtonian 2.4×10^8 - a 67% understatement. The classical ledger fails visibly above $0.5c$.

relativ: labelled physical setup



relativ: limiting curve / diagnostic



REMEMBER / p = gamma m v is the unique momentum that conservation keeps covariant; mass stays invariant.

Why momentum must be gamma m v (source OL5) (cont.)

p = gamma m v from frame-independence; scope fence for classical mechanics; preview of PART 28.

Gamma drills (cont.)

CHECK / γ identity: $\gamma^2(1-v^2/c^2)=1$ - verify at 0.8c: $2.78(0.36)=1.0$.

A cart of mass M_0 (empty) rolls at v_0 on a smooth floor under a hopper that drops sand at rate $\dot{m}$ (kg/s). Collecting mode (source family CM-16/Q22): horizontal momentum is conserved (dropped sand has none): $M(t)v(t)=M_0v_0$ with $M(t)=M_0+\dot{m}t$: $v(t)=\dfrac{M_0v_0}{M_0+\dot{m}t}$. The retarding force at any instant:

Cart under a hopper: the full variable-mass ledger

$F=\dot{m}v$ (the sand must be dragged from 0 to v). Power drained from the cart's KE: exactly $\frac{1}{2}\dot{m}v^2$ - half becomes the sand's KE, half becomes pickup heat (CM-16's factor-2).

Leaking mode: a cart of mass M at v leaks sand vertically at rate $\dot{m}$: the dropped sand carries horizontal momentum $\dot{m}v$ away per second, so $\frac{d}{dt}(Mv)=\dot{m}v$; with $\frac{dM}{dt}=-\dot{m}$ this gives $M\dot{v}=0$: v CONSTANT. The classic surprise: a leaking cart does not slow down; only loading slows it.

Cart under a hopper: the full variable-mass ledger (cont.)

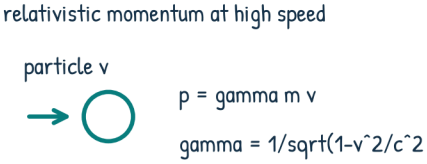
Engine mode: a cart with engine force F collecting sand: $\frac{d}{dt}[(M_0+\dot{m}t)v]=F$; solve the linear ODE. With constant power P instead: $\frac{d}{dt}[\frac{1}{2}Mv^2]=P-\frac{1}{2}\dot{m}v^2$ (the pickup tax). The hopper family is the rocket equation with the sign of $\dot{m}$ flipped and no exhaust speed - same ledger, one diagram.

Cart under a hopper: the full variable-mass ledger (cont.)

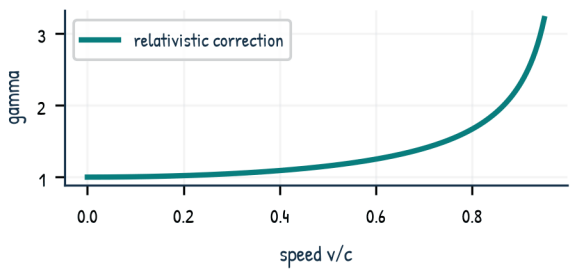
WHY / these three modes (load, leak, thrust) exhaust the first-order variable-mass list for exam mechanics; the momentum ledger in dt is the single method. The surprise results (leaking keeps v; loading burns half the delivered power as heat) are examiner favourites because they punish intuition.

CHECK / collecting numbers: $M_0=20$ kg at 5 m/s, $\dot{m}=0.1$ kg/s, at $t=100$ s: $v=100/30=3.33$ m/s; KE: 250 J to $\frac{1}{2}(30)(11.1)=166.6$ J; the 83.4 J lost is pickup heat, matching the differential identity $\frac{1}{2}\dot{m}v^2=-\frac{d}{dt}KE$ (verified in CM-16).

relativ: labelled physical setup



relativ: limiting curve / diagnostic



REMEMBER / p = gamma m v is the unique momentum that conservation keeps covariant; mass stays invariant.

Cart under a hopper: the full variable-mass ledger

Load/leak/thrust modes; sprinkler and mixed flows; drills with the surprise results.

Sprinkler cart and jet recoil (the same ledger)

Sprinkler: cart + water tank ejects water at $\dot{m}$ with speed v_e relative to the cart (usually backward): thrust $\dot{m}v_e$; the cart is a rocket with water propellant: $\Delta v = v_e \ln(M_0/M_f)$ if $\dot{m}$ runs to dry.

Fire-extinguisher recoil: treat the spray as exhaust. A cart under a leaky bucket with a pump: combine loading and thrust terms in one dt equation.

Sprinkler cart and jet recoil (the same ledger) (cont.)

Mixed problem: cart collects rain at $\dot{m}_1$ while ejecting at $\dot{m}_2$ with relative speed v_e: the systematic form is $M\dot{v} = F_{ext} + \dot{m}_{in}(u_{in} - v) + \dot{m}_{out}(u_{out} - v)$ with all velocities lab-frame; the CM-16 sign discipline sets each term.

CHECK / sprinkler limit: no water: $\Delta v = 0$; no external force with ejecta: momentum of cart + ejecta conserved exactly.

Hopper drills

(a) Cart 30 kg at 6 m/s collects 2 kg of dropped sand: $v = 180/32 = 5.625$ m/s. (b) Cart 30 kg at 6 m/s LEAKS 2 kg vertically: $v = 6$ m/s unchanged. (c) Cart 30 kg at rest ejects 5 kg backward at 4 m/s relative to the cart: ejected lab speed $= v - 4$; momentum $0 = 25v + 5(v-4) = 30v - 20$: $v = 0.67$ m/s.

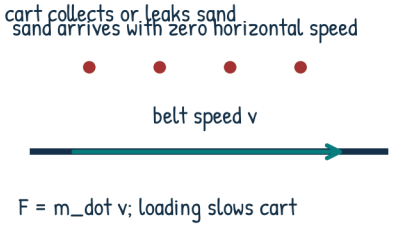
(d) Engine force 60 N while loading at 0.2 kg/s (cart 30 kg at 6 m/s initially): $(30 + 0.2t)v = 180 + 60t$; at $t = 10$: $32v = 780$: $v = 24.4$ m/s.

Hopper drills (cont.)

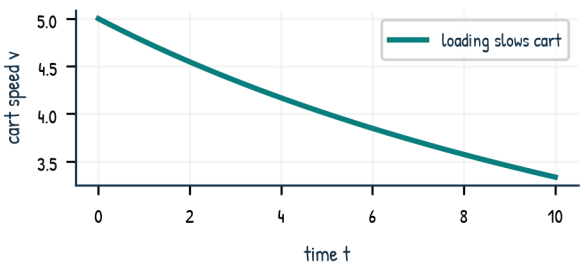
WATCH / Trap: forgetting the $\dot{m}v$ retarding term when loading. Trap: assuming leaking slows the cart. Trap: using $\Delta v = v_e \ln(\cdot)$ for a collecting cart (that formula is for ejecting mass).

CHECK / drill (c): $p = 25(0.67) - 5(3.33) = 16.7 - 16.7 = 0$; drill (d): $32 \times 24.4 = 780 = 180 + 600$.

hopper: labelled physical setup



hopper: limiting curve / diagnostic



REMEMBER / Hopper cart: loading slows (F = m_dot v), leaking does not; the d(Mv)/dt ledger solves all three modes.

Cart under a hopper: the full variable-mass ledger (cont.)

Load/leak/thrust modes; sprinkler and mixed flows; drills with the surprise results.

Explicit solutions of the loading-engine ODE

$\frac{d}{dt}[(M_0+\dot{m}t)v]=F$ integrates directly: $(M_0+\dot{m}t)v=M_0v_0+Ft$: $v(t)=\dfrac{M_0v_0+Ft}{M_0+\dot{m}t}$.

Long-time limit: $v\to F/\dot{m}$ (the engine can only hold that asymptote while dragging new mass). With $F=0$ it reduces to the pure-loading decay $M_0v_0/(M_0+\dot{m}t)$; with $\dot{m}=0$ it is uniform acceleration.

Explicit solutions of the loading-engine ODE (cont.)

Drill: $M_0=10$, $v_0=2$, $\dot{m}=0.5$, $F=20$: $v(10)=(20+200)/15=14.7$ m/s; asymptote $F/\dot{m}=40$ m/s. Power at $t=10$: $Fv=294$ W; pickup tax $\frac{1}{2}\dot{m}v^2=54$ W.

CHECK / differentiate the closed form at $t=0$: $\dot{v}=[FM_0-\dot{m}M_0v_0]/M_0^2=(200-10)/100=1.9$ m/s^2; the ODE $M\dot{v}=F-\dot{m}v=20-1=19$ gives $\dot{v}=1.9$ - the two agree.

When momentum needs calculus: the three integration patterns

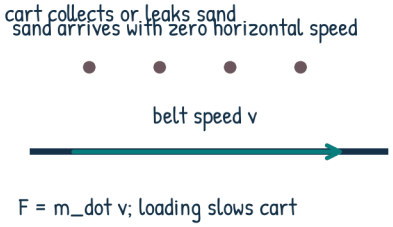
Pattern 1 - VARIABLE FORCE with time: $J=\int F(t)dt=\Delta p$; if F depends on v (drag), write $m\frac{dv}{dt}=F(v)$ and separate: $\frac{m\,dv}{F(v)}=dt$; integrate. Pattern 2 - VARIABLE FORCE with position: $F=F(x)$; use work-energy $\int F dx=\Delta K$ instead (or $v\,dv=a\,dx$).

Pattern 3 - VARIABLE MASS: the dt ledger of CM-16/OLY-10: $\frac{d}{dt}(Mv)=F_{ext}+\dot{m}_{in}u_{in}-\dot{m}_{out}u_{out}$; solve the linear ODE.

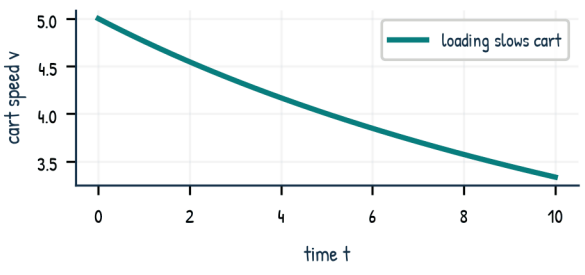
When momentum needs calculus: the three integration patterns (co.

Worked (chain ODE, source OL7 method): $a=g(2x-L)/L$ with $v=dx/dt$: $v\,dv=a\,dx$ integrates to $v^2(x)$; then $dt=dx/v(x)$ integrates to $t(x)$ - the two-integration pattern of every chain problem. Worked (drag): $m\dot{v}=-bv$: $v=v_0e^{-bt/m}$ - exponential relaxation; distance $=\frac{mv_0}{b}(1-e^{-bt/m})$; the body never stops in finite time - a favourite limit question.

hopper: labelled physical setup



hopper: limiting curve / diagnostic



REMEMBER / Hopper cart: loading slows (F = m_dot v), leaking does not; the d(Mv)/dt ledger solves all three modes.

When momentum needs calculus: the three integration patterns

Three integration patterns; separable toolkit; phase-line sketch; drills with the drag and chain ODEs.

When momentum needs calculus: the three integration patterns (co.

Worked (rocket with gravity): $m\dot{v} = -v_e\dot{m} - mg$ with $m = m_0 - \dot{m}t$: $\Delta v = v_e \ln(m_0/m_f) - gt_b$ - the dt ledger with a prescribed $m(t)$. The integration pattern is identical to the hopper engine mode (OLY-10) with sign changes.

WHY / momentum conservation is a first-order statement about p; as soon as the force law or the mass law is nonlinear, the exam problem is really a differential equation. The triage: (i) is F given as t, x, or v? (ii) is mass constant? (iii) choose $J = \int F dt$ or work-energy accordingly.

When momentum needs calculus: the three integration patterns (co.

CHECK / drag limit checks: $b \to 0$: $v \to v_0$; $t \to \infty$: $v \to 0$ with finite distance; large b: stops almost immediately. Chain ODE check: $v^2 = \frac{2g}{L}(x - \frac{L}{2})^2$ differentiated back gives $a = \frac{2g}{L}(x - \frac{L}{2})$ - matches Newton. $mv\frac{dv}{dx} = F(x)$: energy in disguise. $m\frac{dv}{dt} = F(v)$: time-separable. $\frac{d}{dt}(M(t)v) = F(t)$: integrating factor $1/M(t)$. $\frac{d}{dt}(Mv) = \dot{m}u$: variable mass. Recognising WHICH of the four forms a problem takes is the whole exam skill; each has a one-line solution template.

The separable-equation toolkit for mechanics

Phase-line view (olympiad flavour): for $a = a(x)$ the fixed points are $a = 0$ (equilibrium; stable if $da/dx < 0$); the chain over the peg has unstable equilibrium at $x = L/2$; the fold has $a \propto y$ (no fixed point except 0, unstable). Sketching a vs x before integrating prevents sign errors in $v^2(x)$.

The separable-equation toolkit for mechanics (cont.)

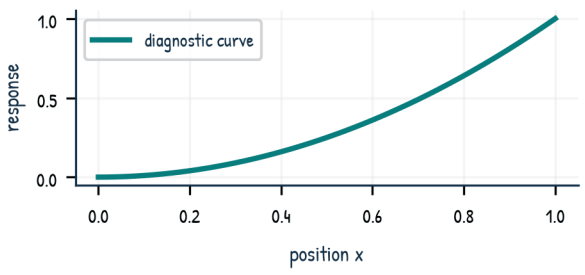
numerical fallback: if the ODE has no closed form (e.g. v^2 drag plus thrust), Euler-step in a table - method marks come from the SETUP $m\dot{v} = \ldots$, and a one-step estimate often answers 'show that the speed exceeds...' questions.

CHECK / toolkit sanity: every template above is just $F = dp/dt$ rearranged; if two templates disagree on a shared problem, the bookkeeping of $M(t)$ vs m is wrong.

integrate: labelled physical setup

integrate
choose system and time window
COM -> definition / symmetry
collision -> momentum + e
flux -> m_dot times delta v
audit signs, energy, limits

integrate: limiting curve / diagnostic



REMEMBER / Three integration patterns: J = int F dt for F(t), work-energy for F(x), the M-v ledger for variable mass.

When momentum needs calculus: the three integration patterns (cont.)

Three integration patterns; separable toolkit; phase-line sketch; drills with the drag and chain ODEs.

Integration drills

(a) $m = 2$ kg, $F = 6t$ N: $\Delta p = \int_0^2 6t\,dt = 12$ N s; $\Delta v = 6$ m/s. (b) $F = -kx$ from $x = 0$ to A: work-energy: $\Delta K = -\frac{1}{2}kA^2$ (spring stop: $\frac{1}{2}mv^2 = \frac{1}{2}kA^2$). (c) Drag bv with $b = 0.4$, $m = 1$, $v_0 = 10$: after 2 s: $v = 10e^{-0.8} = 4.49$ m/s. (d) Chain $L = 1$, $g = 10$, nudge release:

Integration drills (cont.)

$v = \sqrt{20}\,(x - 0.5)$; time from $x = 0.75$ to $x = 1$: $t = \int_{0.75}^{1} \frac{dx}{\sqrt{20}\,(x - 0.5)} = \frac{\ln 2}{\sqrt{20}} = \frac{0.693}{4.472} = 0.155$ s.

WATCH / Trap: using $J = F_{avg}t$ when F is not constant - integrate instead. Trap: mixing $F(x)$ problems into $\int F dt$ (wrong variable). Trap: losing the $\frac{1}{2}$ in $v\,dv = a\,dx$ integration ($v^2/2$, not v^2).

Integration drills (cont.)

CHECK / drill (d) via energy: $v(x) = \sqrt{20}\,(x - 0.5)$ comes from $v^2 = 20(x - 0.5)^2$ (OLY-05); $t = \int dx/v$ = the second integration of Pattern 1.

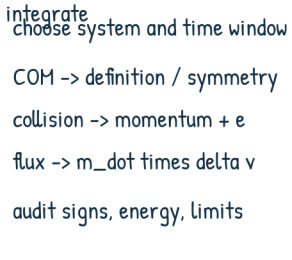
Near an unstable equilibrium, $a \approx \lambda_1(x - x_*)$ with $\lambda_1 > 0$ gives $x - x_* \propto e^{\sqrt{\lambda_1}\,t}$ - the linearised escape law. Peg: $x_* = L/2$, $a = \frac{2g}{L}(x - \frac{L}{2})$: departure rate $\sqrt{2g/L}$; a 1% displacement grows by e in time $\sqrt{L/2g}$. For $L = 1$ m: 0.15 s per e-fold.

Linearise around the fixed point: the exponential departure (cont.)

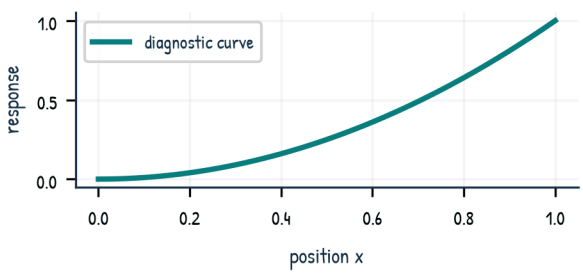
Stable side: if $a = -\omega^2(x - x_*)$, the linearised solution is SHM at ω - the same local analysis classifies every chain/pendulum equilibrium before solving the full ODE.

CHECK / the fold's $a = \frac{2g}{L}y$ is the same linear form with $y_* = 0$: pure exponential from the start; the peg's full solution $v = \sqrt{2g/L}\,(x - L/2)$ is the EXACT finite-amplitude version of the linearised escape - they coincide because the ODE is exactly linear in $(x - L/2)$.

integrate: labelled physical setup



integrate: limiting curve / diagnostic



REMEMBER / Three integration patterns: J = int F dt for F(t), work-energy for F(x), the M-v ledger for variable mass.